
Seasonal odds for water-energy planning: AI-ready teleconnection data and a logistic-referenced residual Transformer

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Abstract

Weeks-to-years prediction of water for energy sits between weather forecasts and century-scale Earth-system projections: long enough that reservoir and generation decisions matter, short enough that year-to-year variability still dominates. In the Tennessee Valley Authority (TVA) region, winter wet/dry and cool/hot regimes shift hydropower inflows and thermoelectric cooling demand, so even categorical seasonal *odds* would be a useful planning product. The problem is hard. Labelled winters are few; month-by-month anomaly regression on teleconnections collapses toward climatology; and operational seasonal systems such as CFSv2 are only a weak *local* guide for categorical wet/dry outcomes. We therefore isolate a first experiment whose purpose is to test whether a documented large-scale ocean–atmosphere state can shift TVA winter odds at all, before any cascade to streamflow or energy models. All learners share one AI-ready pack of teleconnection indices and sea-surface temperature (SST) boxes. A multinomial logistic model is the skill reference. On a blocked test of 15 winters, logistic hit rate and Heidke skill for wet/dry are 0.67 and 0.43, versus 0.47 and 0.11 for October-initialized CFSv2 on the same reanalysis labels. Fully connected, LSTM, and CNN models do not match that reference; a residual temporal Transformer (logistic path plus a constrained attention residual) is strongest among the compared architectures on both wet/dry and cool/hot. Skill is mainly a winter signal and fades beyond about two seasons. The result is evidence that the seasonal-odds question is tractable—not a weeks-to-years hydrology system and not an operational TVA product.

1 Introduction

A challenging but meaningful problem. Water for energy is decided on a weeks-to-years horizon that neither tomorrow’s weather model nor a century-scale Earth-system projection is built to serve. Hydropower and thermoelectric plants in the TVA region depend on water availability, streamflow, and river temperature. A wet versus dry winter changes reservoir inflows; a cool versus hot winter changes heating and cooling load. Those seasonal regimes are tied to large-scale ocean–atmosphere modes: ENSO for North American winter precipitation and temperature [5, 6], and the North Atlantic Oscillation (NAO) and Arctic Oscillation (AO) for much of the eastern U.S. winter temperature variance [7, 8]. If public observations and machine learning can shift the *odds* of those regimes, operators of existing infrastructure gain a planning signal under year-to-year variability—even when the model is not yet a deployed forecast.

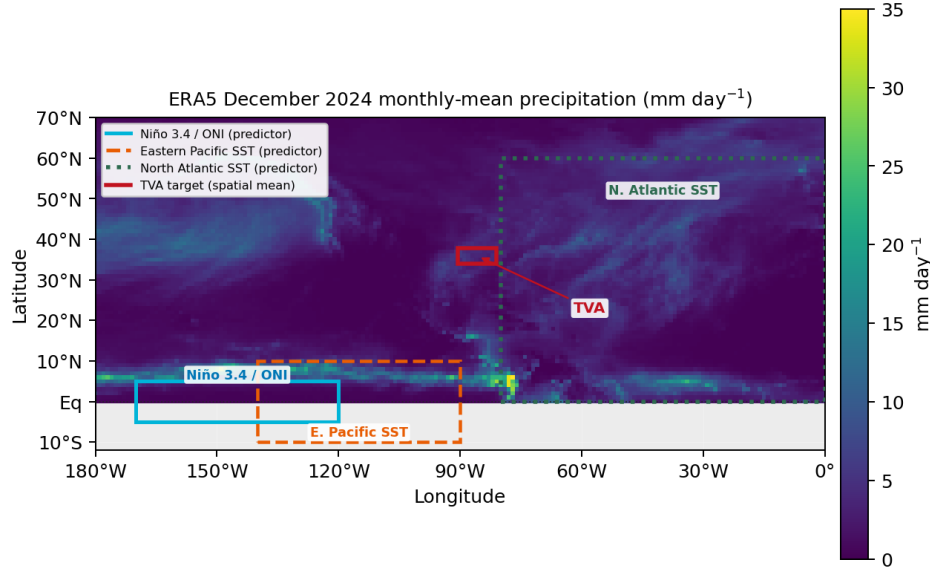


Figure 1: ERA5 December 2024 monthly-mean precipitation (mm day^{-1}) over the daily 1° analysis window, with the index and SST boxes used in this paper. Cyan: Niño 3.4 / ONI; dashed orange: eastern Pacific SST; dotted green: North Atlantic SST; red: TVA spatial-mean target. NAO and AO are atmospheric indices, not additional SST boxes. The southern half of Niño 3.4 lies south of the ERA5 cutoff at the equator (grey); index and SST-box time series still use the full conventional box.

33 The same problem is scientifically hard. The usable sample is small once winters are blocked for
 34 honest evaluation. Pooling every calendar month as a continuous anomaly dilutes the seasonal
 35 signal, so mean-squared-error models look near climatology. Operational seasonal products such
 36 as CFSv2 [9], a member of the North American Multi-Model Ensemble (NMME) [10], need not
 37 be a strong *local* categorical baseline for TVA wet/dry verification. A full cascade from global
 38 atmosphere–ocean fields to streamflow and energy-economy indicators cannot create skill that is
 39 absent in the teleconnection-to-TVA step.

40 **Purpose of this experiment.** We therefore design a constrained first test: can autumn teleconnection
 41 indices and tropical Pacific SST, drawn from one curated pack, shift December–February (DJF)
 42 TVA wet/dry and cool/hot tercile odds? If that signal cannot be recovered with a transparent logistic
 43 model, later spatial, land-runoff, or energy models are unlikely to invent it. The experiment is
 44 evidence-oriented on public data, not an operational TVA system.

45 **How, and what we find.** Figure 1 shows the geography: teleconnection indices and three OISST
 46 boxes as predictors; a TVA ERA5 spatial mean as the target, not map pixels [1–4]. All models see
 47 the same inputs. Logistic regression on September–November (SON) means is the skill reference;
 48 CFSv2 is operational context; a residual Transformer may only refine logistic. On 15 blocked test
 49 winters, logistic already exceeds climatology and this local CFSv2 comparison for wet/dry.

50 2 Related work

51 Teleconnection diagnostics have long linked tropical Pacific SST and North Atlantic pressure patterns
 52 to U.S. winter precipitation and temperature [5–8]. Operational dynamical seasonal prediction—
 53 CFSv2 and NMME—provides a public benchmark [9–11], but global ENSO skill does not automati-
 54 cally transfer to categorical wet/dry verification on a regional energy box. Machine learning has been
 55 applied to ENSO prediction, including deep networks for multi-year ENSO [12]. We instead score
 56 TVA winter *odds* with a logistic reference so a Transformer is accepted only if it matches or improves

that reference on the same curated inputs [13–15]. We train and score on ERA5 TVA means [1] and treat Daymet [17] as a later dual-truth path, not this paper’s target.

3 Data

Every model in this paper—logistic, fully connected (FC) network, LSTM, CNN, residual Transformer—and the CFSv2 categorical comparison use the same public streams: monthly teleconnection indices, including the Oceanic Niño Index (ONI) and Niño 3.4 [3, 4]; SST box anomalies from NOAA OISST [2]; and TVA-mean ERA5 anomalies of near-surface air temperature, precipitation, and mean sea-level pressure [1]. These streams are issued as a monthly labelled pack: one sample per forecast issue month ($N=486$ months, August 1982–January 2023) with a 12-month lookback. The experiments below do not train a second dataset; they aggregate the same series into seasonal means or terciles.

The predictand is ERA5-based, not Daymet. Daily 1° fields over the precipitation window of Fig. 1 are averaged over the TVA box, converted to monthly anomalies, then to DJF terciles (below-normal, near-normal, above-normal) with edges estimated on training winters only. Logistic regression uses SON *means* of the index/SST series; sequence models use the last $L=12$ months of the same series. Reserved dataset DOIs and code URLs appear in the camera-ready data statement.

4 Methods

Tasks. *Winter wet/dry* (primary): SON ONI and Niño 3.4 box anomaly predict the DJF TVA precipitation tercile (dry / near / wet). *Winter cool/hot*: SON NAO and AO predict the DJF near-surface temperature tercile (cool / near / hot).

Split. Train on DJF years ≤ 2007 ($n=25$); test on ≥ 2009 ($n=15$); hold out 2008. Neural models use a five-seed soft-vote ensemble on test.

Scores. Hit rate is the fraction of test winters assigned the correct tercile (chance $\approx 1/3$). Heidke skill measures categorical skill beyond chance ($0 \approx$ chance) [13, 15]. Brier skill score (BSS) is relative to climatological class frequencies [14, 15]; $BSS > 0$ improves on climatology as a probability forecast. These scores match a planner-facing *odds* product rather than pixel RMSE. We say a model *matches the logistic reference* when hit, Heidke, and BSS are all at least as large as logistic on that task.

Models. Multinomial logistic regression on SON means; an FC network on SON means; LSTM and CNN on $L=12$ monthly sequences; and a residual temporal Transformer [16]

$$\text{logits} = \text{logistic}(\text{SON mean}) + \text{softplus}(\alpha) \text{Transformer}(L=12). \quad (1)$$

With ~ 25 training winters, a pure sequence model must rediscover the SON-mean teleconnection from scratch and, in our trials, failed the wet/dry task. The residual form keeps the logistic path (warm-started on SON means) and lets attention learn only a constrained correction.

5 Results

Table 1 reports blocked test scores.

October-initialized CFSv2 ensemble-mean precipitation is mapped to the TVA box and scored as DJF terciles against the **same ERA5 wet/dry labels**. It exceeds the majority-class null (hit 0.47 versus 0.33) but remains weakly skillful categorically. Logistic regression on two autumn means is substantially stronger (hit 0.67, Heidke 0.43). CFSv2 is therefore context, not the reference a learning model must beat. Cool/hot is weaker (logistic hit 0.40). Only the residual Transformer matches logistic on all three metrics for both tasks; “strongest” means strongest among this closed comparison, and the winning model still contains a logistic branch.

Table 1: Blocked test scores on identical AI-ready predictors. CFSv2 is shown for wet/dry as operational context only.

Task	Model	Hit	Heidke	BSS	vs logistic
Wet/dry	CFSv2 Oct→DJF	0.467	0.111	—	below
Wet/dry	Logistic	0.667	0.432	0.091	reference
Wet/dry	FC	0.667	0.432	0.051	below (BSS)
Wet/dry	LSTM	0.400	0.100	−0.016	fail
Wet/dry	CNN	0.333	0.057	−0.038	fail
Wet/dry	Residual Transformer	0.667	0.432	0.140	matches
Cool/hot	Logistic	0.400	0.167	0.009	reference
Cool/hot	FC	0.333	0.112	0.043	partial
Cool/hot	LSTM	0.267	0.046	0.037	partial
Cool/hot	CNN	0.267	0.000	0.022	partial
Cool/hot	Residual Transformer	0.467	0.195	0.050	matches

6 Discussion and limitations

Two extra checks bound the claim. Using the same winter-style indices on other meteorological seasons, only autumn wet/dry also clears a weak pass; spring and summer do not (Appendix A). Lengthening lead while keeping winter wet/dry as the target, a one-season issue (SON→DJF) reproduces the main result; two seasons (JJA→DJF) is a weak pass (hit 0.40, BSS +0.024); three seasons (MAM→DJF) fails. A residual Transformer does not match logistic at the two-season lead. Useful index horizon is therefore about one to two seasons for winter wet/dry, not three seasons and not all calendars.

The test set contains only 15 winters, so scores are a pilot demonstration rather than an operational skill claim. Cool/hot skill is modest; the water-energy narrative is primarily wet/dry. The residual Transformer is not trained from a blank slate. CFSv2 is evaluated here for TVA-box categorical wet/dry odds, not as a global seasonal verification package [11]. This work does not yet forecast streamflow, river temperature, or energy-economy outcomes; those later steps should reuse the same odds language once daily or land targets exist.

7 Conclusion

Weeks-to-years water-for-energy planning needs seasonal TVA information that is difficult but, on AI-ready teleconnection data, tractable as winter odds. Logistic regression already exceeds climatology and this local CFSv2 categorical comparison for wet/dry; a residual Transformer is the strongest model we compared, with logistic retained as the reference. That is evidence for a ground-up seasonal water-energy step—not a weeks-to-years hydrology system, not an image downscaler, and not a deployed TVA product. Next data work is a daily or weekly pack, not further architecture search on the monthly index tensor.

A Calendar transfer and longer leads

Table 2 uses the same logistic recipe with a strict prior-season lag (train season-years ≤ 2007 , test ≥ 2009). Table 3 keeps winter wet/dry and issues the forecast one, two, or three seasons ahead.

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Table 2: Logistic skill by meteorological season (strict prior-season predictors). A pass requires BSS > 0 and (Heidke > 0 or hit above the majority class).

Season	Task	Hit	Heidke	BSS	Pass
DJF (winter)	wet/dry	0.667	0.432	0.091	yes
DJF (winter)	cool/hot	0.400	0.167	0.009	yes
MAM (spring)	wet/dry	0.333	−0.304	−0.207	no
MAM (spring)	cool/hot	0.133	−0.016	−0.122	no
JJA (summer)	wet/dry	0.400	0.075	−0.011	no
JJA (summer)	cool/hot	0.333	0.063	−0.091	no
SON (autumn)	wet/dry	0.400	−0.125	0.016	yes
SON (autumn)	cool/hot	0.400	−0.063	−0.295	no

Table 3: Logistic winter wet/dry skill versus issue lead (cool/hot at two seasons shown for contrast).

Issue → target	Hit	Heidke	BSS	Pass
SON→DJF (one season)	0.667	0.432	0.091	yes
JJA→DJF (two seasons)	0.400	0.135	0.024	yes
MAM→DJF (three seasons)	0.133	−0.121	−0.230	no
JJA→DJF cool/hot	0.267	−0.071	−0.084	no

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